

Space Talk

Visualizing how outer space has been framed, addressed,
and imagined over seven decades

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Abstract

Outer space is a domain as symbolic as it is physical. From the Cold War space race to the rise of commercial spaceflight, the material history of space exploration has unfolded alongside a parallel history of narrative: how space has been framed, addressed, and imagined. This project maps that narrative history across seven decades through two parallel records. The first uses semantic embeddings to analyze news headlines and presidential communication and trace how outer space has been framed in elite mainstream media coverage and presidential rhetoric in the United States, identifying recurring narrative frames and actor types over time. Alongside this discursive analysis of talking about space runs its counterpart: a record of talking *to* space, visualized through a catalog of cultural artifacts and messages sent beyond Earth, drawing from research by Quast and Dunér (2024).

This hybrid approach examines outer space as a political and cultural object, exploring what its construction, decade by decade, reveals about the societies that looked up at it.

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1. Introduction

In 1957, Sputnik's radio signal beeped every 96 minutes for 21 days, orbiting Earth uncontested. In 1977, NASA engraved the sound of a kiss on a golden record and launched it into interstellar space. In 1986, seven astronauts died on live television as a space shuttle disintegrated 73 seconds after liftoff. In 2021, a seat on a private rocket sold at auction for more than twenty-eight million dollars.

Outer space is a domain at once physical and symbolic—one that different eras, institutions, and actors have approached through different logics: national prestige, scientific curiosity, existential risk, commercial opportunity, and something closer to existential awe. This project explores how those logics have shifted over time, across the different arenas in which space is publicly constructed. To do so, it traces how outer space has been discursively framed across two key domains from the 1950s to the present: media coverage and presidential rhetoric.

The space age, inaugurated with Sputnik in 1957, is less a single coherent era than a succession of distinct cultural moments. Its initial phase was defined by Cold War nationalism, with space as a battlefield for ideological legitimacy as much as technical achievement. The Apollo program exemplifies this logic: a state-led initiative framed primarily through geopolitical competition, even as it was simultaneously narrated through the language of universal human achievement. Following the Cold War, while tensions persisted, space narratives in the United States shifted toward pragmatism and internationalism, illustrated by the International Space Station. Most recently, the rise of commercial space actors has introduced an entrepreneurial reframing, positioning space as a frontier for markets, private ambition, and resource extraction (Ward 2019; Nam 2023).

Scholarship on space as a cultural and political object draws from several fields. Space has been examined as a constitutive element of sociotechnical imaginaries (Jasanoff and Kim 2009) and as an arena of geopolitical competition and power projection (Dolman 2002; Siddiqi 2010). The role of media and communication in shaping public support for space programs has also received sustained attention (Kauffman 1994), as has the more recent privatization of spaceflight and its consequences in narratives surrounding this arena (Ward 2019; Nam 2023). This project sits at the intersection of these traditions, contributing a data-centric approach to questions that have largely been addressed through qualitative historical analysis.

The two discursive layers examined are articles from *The New York Times* as a proxy for elite mainstream media coverage, and presidential communication archives from the American Presidency Project as a record of institutional political framing. Each layer is examined to analyze the dominant themes and actors across time, enabling systematic cross-domain comparison.

Alongside this approach, as a distinct, complementary layer, is a timeline of cultural artifacts and messages dispatched into space, collected and visualized as material evidence of how humanity has chosen to project outward into space, from interstellar radio messages, to time capsules, to

commercial advertisements (Quast and Dunér 2024). This way Space Talk seeks to show both sides of space framing: where the discursive layers examine how space is narrated for audiences on Earth, this strand asks what humanity has chosen to project outward.

2. Treatment

2.1. Space as a Discursive Object

The launch of Sputnik 1 on October 4, 1957 is the conventional starting point for the space age. It marked humanity's first sustained material engagement with outer space: an object built by human hands, circling the Earth, beeping. But Sputnik was also a media event. American radio stations broadcast its signal and newspapers showed the story on front pages. The Soviet achievement was immediately visible not only as a scientific milestone, but as a projection of technological and political power (Dickson 2001).

This fusion of a concrete technological event with media spectacle is constitutive of the history of space exploration, an endeavor that has, from the beginning, been showcased publicly. The space age produced not only scientific and technological landmarks but a massive volume of discourse—news coverage, political speeches, fiction, advertising—through which societies made sense of what was happening and what it meant.

Scholars have paid particular attention to the importance of media and communication strategies in the construction of public support for space programs. NASA's relationship with *Life* magazine, for instance, granted exclusive access to astronauts' personal lives in exchange for coverage intended to generate public engagement (Kauffman 1994). The moon landing of 1969 was watched live by an estimated 125–150 million Americans (Teichner 2019). The space age was, in this sense, a media phenomenon as much as a scientific one, and its public meanings were always partly constructed.

The space age also generated its own analytical frames. The language of the “race” framed space exploration as a zero-sum competition between political opponents. The language of the “frontier” drew on American mythologies of westward expansion to position space as the next chapter of national destiny: appeals to Columbus, European mariners, the settlement of the American West (McCurdy 1997). These frames shaped public understanding of space in significant ways.

As the superpower rivalry that had driven Apollo-era investments during the Cold War receded, space policy in the United States reoriented toward pragmatism and international cooperation. The International Space Station exemplified this post-Cold War logic: as Sadeh (2004) argues, the program evolved through successive phases of coordination, augmentation, and interdependence, each shaped by changing domestic and international political conditions. At the same time, the Challenger disaster of 1986 and the Columbia disaster of 2003 exposed a counter-narrative to the triumphalist story of technological progress, signaling the fragility and institutional failure underlying space endeavors (Hall 2016).

In recent years, the rise of commercial space endeavors has transformed the sector's ecosystem, with public agencies progressively delegating the direction of innovation to private actors and market

forces (Robinson and Mazzucato 2019). The emergence of space technology companies such as SpaceX has introduced a new set of actors, a new vocabulary, and new framings of what space is for. Here, commercial opportunity refers not only to highly visible ventures such as space tourism, but more fundamentally to the expansion of routine, revenue-generating activities, such as satellite deployment for communications and broadband, Earth observation for agriculture and climate monitoring, and launch services that increasingly function as critical infrastructure for the global data economy. A clear example of this emerging and increasingly centralized commercial infrastructure is the rapid expansion of the Starlink constellation, which has already deployed more than 10,000 satellites and now constitutes a dominant share of low-Earth-orbit capacity for satellite internet (Shepardson 2026). At the same time, entrepreneurs have reframed space through the language of markets, private ambition, and resource extraction, positioning colonization of other planets as both a commercial opportunity and a civilizational imperative (Ward 2019; Nam 2023). This shift from publicly coordinated scientific endeavor to a system increasingly structured around markets and long-term species survival narratives has provoked debate about whether the privatization of space represents a fundamental reorientation of its cultural logic from a public good coordinated by states to a frontier shaped by capital (Robinson and Mazzucato 2019; Nam 2023).

2.2. Data Sources

2.2.1. Talking About Space

This project analyzes how outer space has been narrated across two distinct discursive layers, each capturing a different register and audience. Together, they form a composite picture of how space has been publicly constructed in the United States from the 1950s to the present.

Elite mainstream media coverage: The New York Times. The first layer draws on articles from The New York Times, collected via the NYT Article Search API. The Times is used as a proxy for elite mainstream media coverage for several reasons: its continuous publication record across the entire period of interest, its historical role in setting the agenda for broader national coverage, and the availability of a structured, queryable archive. Articles were retrieved using a set of twenty-five high-precision search phrases spanning major frames of space discourse, from “space race” and “moon landing” to “anti-satellite” and “dark matter” (see Annex A for the complete list). Only articles where the phrase appeared in the headline were retained, ensuring that the search term reflects the article’s primary theme rather than an incidental mention. After deduplication and cleaning (including the manual removal of false positives such as articles about the Sputnik V COVID-19 vaccine), the corpus comprises 6,442 headlines covering 1950–2024.

Presidential rhetoric: The American Presidency Project. The second layer draws on key documents—speeches, press conferences, signing statements, briefings—collected from the American Presidency Project at the University of California, Santa Barbara. Rather than analyzing full documents, which may mention space only in passing, the pipeline extracted space-relevant excerpts: the exact paragraphs in which any search phrase appeared, with no surrounding context added. Where a single document matched multiple search phrases in different paragraphs, all matching paragraphs were merged into one excerpt and deduplicated, so the unit fed to the classifier is all space-relevant passages from that document. Upon review, certain entries were supplemented with related passages from the same document to ensure that all relevant context was preserved—particularly in cases where space was discussed within a Q&A format, and including both the question and the answer provided a more complete and meaningful interpretation. This excerpt-based approach ensures that every analyzed unit is genuinely about space, rather than a document that names space among many other topics. The presidential corpus comprises 1,408 excerpts covering 1960–2024, with the 1950s excluded from the analysis due to insufficient data.

Because the corpora are generated through a high-precision keyword strategy, the resulting analysis should be interpreted as patterns within a deliberately constructed corpus of space-related discourse. The search terms inevitably shape the archive. The analysis therefore does not measure the absolute prevalence of space narratives in U.S. public discourse; instead, it compares how a defined corpus of explicitly space-related texts is structured across sources and decades.

2.2.2. Talking to Space

This layer offers a qualitative, object-level record that complements and contextualizes the aggregate patterns surfaced by the discourse analysis. Where the previous approach asks how space is narrated inward, this section asks what we have chosen to project outward.

Since the beginning of the space age, humanity has dispatched into space not only scientific and technological instruments but also cultural artifacts and messages. These range from interstellar radio transmissions, to time capsules, to advertisements. For the purposes of this project, a database is drawn directly from the taxonomic catalogue assembled in *Speaking Beyond Earth: Perspectives on Messaging Across Deep Space and Cosmic Time* (Quast and Dunér, 2024), which provides one of the most comprehensive scholarly inventories of cultural objects, messages, and transmissions sent to outer space.

The catalogue was treated as a curated qualitative dataset rather than as an exhaustive statistical corpus. Each entry was coded by year, object/message type, initiating actor, and category. The visualization therefore emphasizes historical sequence and changing composition rather than frequency in a representative population.

2.3. Data Processing and Classification

2.3.1. The Embedding Approach

As an initial, exploratory approach, the theme and actor labeling was done using regular expression matching: a predefined list of terms was assigned to each category, and any document containing those terms was flagged accordingly. This approach, however, raises two fundamental limitations when working at scale. First, vocabulary mismatch: a document about rivalry might use terms such as “competition,” “threat,” or “falling behind” rather than the specific word “rivalry.” Extending the keyword list to cover all possible variants becomes an unmanageable iterative exercise in enumeration with no clear end. Second, semantic ambiguity. Certain terms carry multiple meanings depending on context. For instance, the word “Sputnik” appears in headlines about the 1957 Soviet satellite but also—as surfaced during data collection—in headlines about the Sputnik V COVID-19 vaccine. Regular expressions cannot distinguish these without increasingly complex processes that are imperfect and specific to each corpus. Sentence embeddings help address both problems by representing meaning as a numerical position in a high-dimensional space, so that documents with similar meaning are geometrically close regardless of whether they share specific words.

Every document (each headline and political excerpt) and every theme and actor anchor is converted into a 768-dimensional numerical vector using *nomic-embed-text-v1.5*, an open-source sentence embedding model. The position of a vector in this 768-dimensional space encodes meaning: documents that say semantically similar things end up geometrically close to each other. Cosine similarity (the angle between two vectors) is then used to measure how close each document is to each theme and actor anchor.

A feature of the model relevant to this setup is its task instruction prefix system. Theme and actor anchors are embedded with the prefix “search_query:”, while documents are embedded with the prefix “search_document:”. These prefixes, specified in the model’s documentation as the intended usage pattern for retrieval tasks, ensure that the embeddings are calibrated for asymmetric comparison: the anchor expresses “what I am looking for” and the document expresses “what this text is about,” rather than both being treated as equivalent texts being compared side by side.

For NYT headlines, the input for the classification model is one headline. For presidential excerpts, the input is the pre-extracted excerpt: the merged, deduplicated set of space-relevant paragraphs from each document. The pipeline uses chunked-max pooling. Each excerpt is passed through the same chunking function: short excerpts produce a single chunk, while longer excerpts are split into overlapping 140-word chunks with a 90-word stride, creating a 50-word overlap between consecutive chunks. The overlap exists so that a sentence sitting at the boundary between chunks is fully represented in at least one of them, rather than being cut in half. Each chunk is embedded

independently. For each theme, the excerpt-level score is then defined as the maximum cosine similarity obtained by any chunk for that theme, rather than the average across chunks. Max pooling captures whether a theme is strongly present anywhere in the excerpt: if one passage is strongly about rivalry while the rest references other subjects, averaging would dilute that signal.

2.3.2. Thematic Framing

Both corpora are analyzed through a shared taxonomy of six thematic frames, developed iteratively to capture the major registers in which space discourse operates. The frames are:

- Space race & rivalry: geopolitical competition, threat perception, military use of space, superpower rivalry
- Spaceflight & missions: the operational, physical reality of being in space (launches, missions, crews, NASA planning)
- Studying the cosmos: scientific discovery and observation (dark matter, exoplanets, observatories)
- Risk & hazards: failure, danger, and disasters
- Markets & commerce: commercial spaceflight, private sector actors, space tourism, and the entrepreneurial reframing of space
- Meaning & identity: philosophical and existential dimensions (what space means for humanity, symbolic implications)

Each frame is represented by a dense natural-language anchor query, each one a multi-sentence description of what could constitute a document belonging to that theme across the time frame analyzed (See Annex B for full theme queries.) For example, the rivalry frame anchor explicitly references both Cold War rivals (USSR, Sputnik) and contemporary ones (China, CNSA, anti-satellite weapons.)

Prior to classification, anchor distinctiveness was validated by computing pairwise cosine similarities between all theme anchors. The threshold used was 0.90: above this value, two anchors would not be distinguishable enough to the model, producing unreliable discrimination. The resulting matrix is shown in Annex D.

Cosine similarity is computed between each document's embedding and each of the theme anchors, producing a similarity vector per document. However, because all documents in the corpus are space-related, they tend to be relatively similar to all space-related anchors. As a result, raw cosine similarity alone presents limitations for drawing conclusive links. The critical step is

z-score normalization. For each document, the six similarity scores are standardized relative to that document's own mean and standard deviation. This reframes the task: rather than asking "which theme is this document closest to?", it asks "which theme does this document stand out in, relative to its similarity to all other themes?"

Softmax is then applied to the z-scores, exponentiating each value and normalizing by their sum to produce a relative distribution of theme weights that sums to 1.0.

This is how the process works in the example headline "NASA Had Solution to Key Flaw in Rocket When Shuttle Exploded."

Step 1 — Raw cosine similarities

Theme	Raw similarity
Risk & hazards	0.744
Spaceflight & missions	0.685
Space race & rivalry	0.645
Studying the cosmos	0.613
Markets & commerce	0.618
Meaning & identity	0.540

Step 2 — Z-score normalization

Each score is transformed as $(\text{score} - \text{mean}) / \text{SD}$:

Theme	Raw similarity	Z-score
Risk & hazards	0.7440	+1.63
Spaceflight & missions	0.6849	+0.70
Space race & rivalry	0.6447	+0.06
Studying the cosmos	0.6128	−0.44
Markets & commerce	0.6177	−0.36
Meaning & identity	0.5403	−1.59

Risk & hazards is now +1.63 standard deviations above this document’s mean. Meaning & identity is −1.59 standard deviations below. The three themes above the document mean (positive z-scores) are “Risk & hazards,” “Spaceflight & missions,” and “Space race & rivalry”.

Step 3 — Softmax

Softmax exponentiates each z-score and divides by the sum, producing a relative distribution of theme weights that sums to 1.0:

Theme	Relative weight
Risk & hazards	0.526
Spaceflight & missions	0.206
Space race & rivalry	0.109
Studying the cosmos	0.066
Markets & commerce	0.072
Meaning & identity	0.021

A relatively small difference in raw similarity (≈ 0.06) translates into a clear dominant assignment after normalization: “Risk & hazards” receives 52.6% of the weight, while “Spaceflight & missions” receives 20.6%, with the remaining themes accounting for the rest.

Robustness check: z-score versus excess weight

As a robustness check, an alternative normalization method was also computed and compared against z-score across all decades: excess weight.

Excess weighting operates differently from z-score normalization. For each document, it subtracts the mean similarity across themes, sets negative values to zero, and renormalizes the remaining positive values. This produces sparser assignments: only themes that exceed the document average receive positive weight. By contrast, z-score softmax preserves graded information across all themes, assigning small but nonzero weights even to those below the mean. Despite these differences, both methods identify the same top two themes in every decade across both corpora. In two decades (2000 and 2010 in the presidential corpus), the ordering of these top two themes is reversed, but

the difference between them falls within a narrow margin. This convergence suggests that the results are not driven by the choice of normalization method, but instead reflect an underlying structure in the data.

2.3.3. Actor Classification

In addition to thematic framing, the corpora is analyzed through an actor taxonomy that identifies which entities are positioned as central to each narrative. The classification process follows the same pipeline as for themes: cosine similarity against actor anchor queries, z-score normalization, and softmax weighting. But two things differ from the theme pipeline: the actor sets are not the same across corpora, and one category (astronauts) is defined by a regular expression rather than embeddings. Because the actor taxonomies are source-specific, actor rankings should be interpreted within each corpus rather than as direct one-to-one comparisons across corpora.

Differing set of actors across corpora

The initial design applied the same five actor categories to both NYT headlines and presidential excerpts: national state, geopolitical rivals, international partnerships, private sector, and scientific community. However, two categories proved unstable and were revised after reviewing classification outputs.

The “international partnerships” category did not perform well in the news corpus. It functioned as a diffuse catch-all, grouping together heterogeneous headlines without a coherent actor structure. This makes sense structurally: news headlines rarely foreground alliance frameworks or partner institutions as primary actors in the way political speech does. Removing this category from the NYT corpus sharpened the remaining actor distinctions.

A parallel problem emerged with the “scientific community” actor type in the presidential communication corpus. Manual review showed that while scientific discoveries are frequently referenced, scientists themselves are rarely explicitly positioned as key actors. Instead, they are invoked to legitimize policy decisions made by state actors. Therefore, the “scientific community” category was retained for the news corpus—where journalists often foreground researchers and institutions—but excluded from the presidential corpus.

The astronaut exception

The astronaut category is the only one assigned by regular expression rather than embeddings. The regex matches only the literal words “astronaut,” “astronauts,” “cosmonaut,” or “cosmonauts” in

the headline. Where it matches, the embedding-based score for this actor is overridden to ensure it dominates; where it does not, the category is excluded entirely.

This adjustment was motivated by an empirical observation. In reviewing the initial classification outputs for the news corpus, a recurring pattern of misclassification emerged: a significant number of headlines were being assigned to institutional actors in ways that did not reflect their narrative focus. Closer inspection revealed a common thread: in many cases, these were astronaut-centered headlines.

This misalignment is also historically intuitive. Astronauts were not merely instruments of state programs, but public figures in their own right: the human face of the space age, heroes. However, embedding-based methods cannot reliably distinguish “a headline about what an astronaut did” from “a headline about what an institution did,” given the strong semantic overlap between the two. This reflects a conceptual distinction between institutional actors and symbolic human figures that is not captured by embedding similarity alone. The solution is therefore to bypass the embedding for this specific category and use a precise rule that identifies when the individual human actor is foregrounded. This correction also improves the remaining categories: removing astronaut-centered headlines reduces spillovers and results in cleaner actor assignments in general.

The astronaut category is not used in the presidential corpus, where astronauts appear less frequently and are typically framed as instruments of institutional programs rather than as independent rhetorical actors.

This rule also privileges precision over recall: it captures explicit astronaut/cosmonaut headlines but does not include specific astronaut names that do not use those terms.

The final actor sets are:

NYT corpus: astronauts, national state, geopolitical rivals, private sector, scientific community.

Presidential corpus: national state; geopolitical rivals; international partnerships, private sector.

Full anchor queries for each actor are in Annex C. Anchor distinctiveness was validated by computing pairwise cosine similarities between all actor anchors (Annex D.)

2.3.4. Sampling for Accuracy

To assess classification accuracy, a manual audit was conducted on the top 30 highest-confidence documents for each actor and theme in each corpus. Rather than drawing a random sample, this approach evaluates whether the center of each category (the documents most strongly associated with it) is correctly identified. As such, this evaluation should be interpreted as a measure of precision at the center, rather than overall accuracy. Future validation could supplement this audit with a stratified random sample across decades and confidence levels.

Actor classification performs consistently well across both corpora, with precision in a high-confidence sample at around 96%. This suggests that the actor taxonomy is well specified and that the model reliably identifies the central entity in each document. Theme classification also performs strongly in the news corpus (92.8%), where language is more literal and event-driven. Performance is lower for thematic classification in the presidential corpus (79.6%), reflecting the more abstract and rhetorical nature of political language. In these cases, themes are often implicit, overlapping, or expressed metaphorically, making precise classification inherently more difficult.

Taken together, these results indicate that when the model assigns high weight to a category, it is usually correct. However, given that the precision varies among sources—with the political theme results on the lower range—the outputs are meant to be interpreted as broad temporal tendencies rather than as precise document-level classifications.

Average precision across top-30 high-confidence samples for each actor and theme category within each source

Category Type	Source	Precision in high-confidence sample
Actor	NYT	96.0%
Actor	Presidential	95.8%
Theme	NYT	92.8%
Theme	Presidential	79.6%

2.4. The Visualization

The project takes the form of an interactive website structured around a landing page and two primary sections. The landing page introduces the central premise and establishes the conceptual frame before presenting the two entry points: *Talking about Space* and *Talking to Space*.

Both main sections begin with scroll-driven narrative sequences, which provide editorial framing introducing key ideas and examples before the data-driven interactive components appear. The interaction design is intended to move from editorial framing to exploratory comparison: the scroll sequences establish the interpretive frame, while the toggles and drilldowns allow users to compare sources, decades, actors, and themes without leaving the narrative structure.

The interactive project is available at: <https://lara-yeyati-preiss.github.io/space-talk/>

2.4.1. Talking about Space

The *Talking about Space* section presents the results of the thematic and actor analyses through two different views: a rank-based visualization of actors and a distribution-based visualization of themes. A source toggle allows users to switch between the NYT and presidential corpora.

The main characters: actor rankings over time

The first view visualizes actor prominence using a rank bump chart. Each actor is represented by a continuous line whose vertical position corresponds to its rank within each decade, computed from the mean softmax weight of that actor across all documents in that period. Because the model assigns each document a normalized set of actor weights, this representation captures the average relative prominence of each actor across all documents in a decade.

The chart is designed to emphasize structural transitions. Line crossings signal changes in narrative dominance, and these crossings anchor the interpretation of the visualization. Annotations highlight key historical events associated with each actor, linking changes in rank to concrete landmarks.

The resulting pattern in the news corpus follows a clear sequence: early dominance by geopolitical rivals, followed by the rise of astronauts during the Apollo era, a shift toward institutional actors during the Shuttle period, and finally the ascent of the private sector in the 2010s and 2020s. In contrast, the political corpus remains historically centered on the national state, with international partnerships becoming more prominent in recent decades. The divergence between these trajectories reflects fundamental differences in how space is framed across journalistic and political discourse.

Figure 1. Actor rankings over time in the New York Times corpus

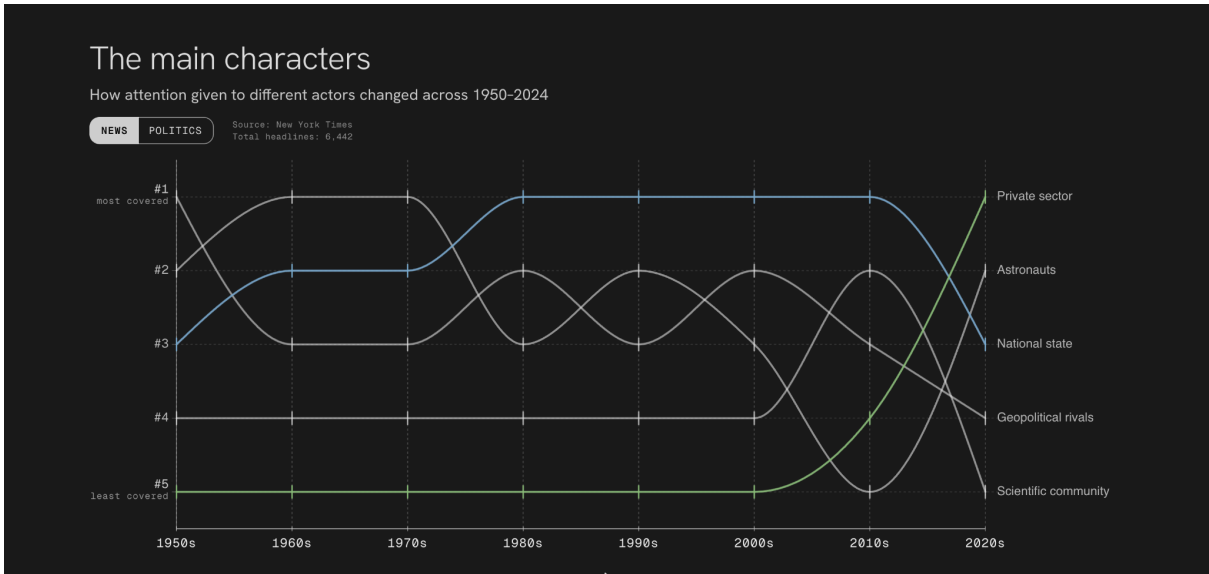
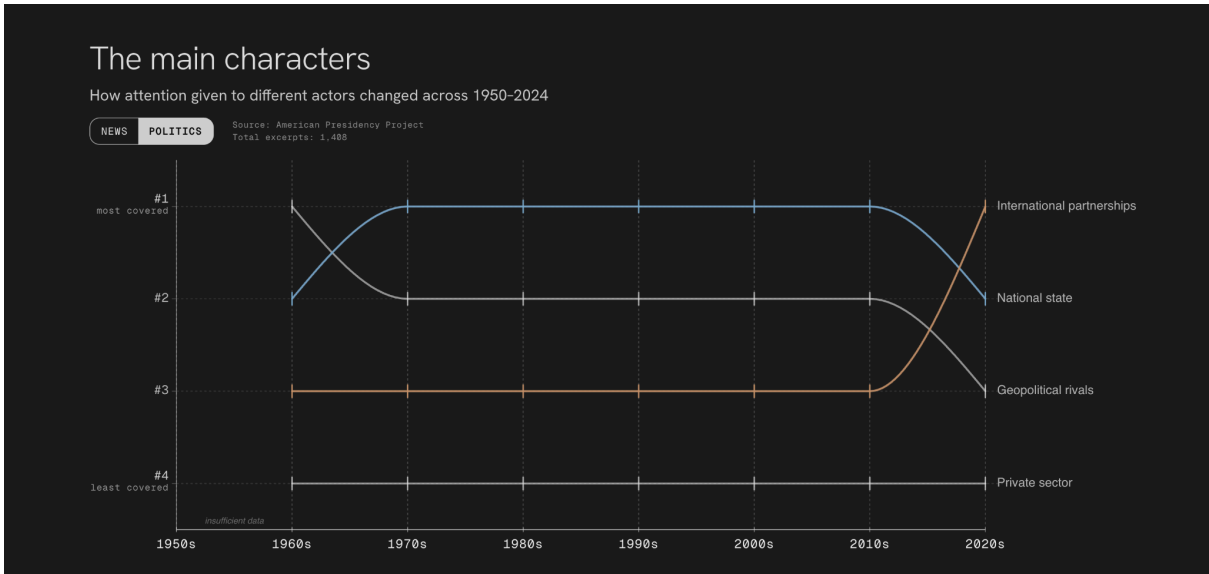


Figure 2. Actor rankings over time in the presidential corpus.



The recurring themes: distribution and evolution

The second view presents thematic structure across two levels: an aggregate overview and a decade-level drilldown. The overview displays one circle per thematic frame, with circle area encoding the mean softmax weight of that theme across the full corpus.

Themes are connected by lines representing pairwise co-occurrence. Co-occurrence is defined as the proportion of documents in which both themes exceed the document's average similarity across all themes. This threshold identifies cases where more than one theme is meaningfully present, rather than merely co-occurring due to background similarity.

Co-occurrence patterns diverge across corpora. In news coverage, “Spaceflight & missions” most strongly overlaps with “Risk & hazards,” with “Space race & rivalry” being the secondary connection. In political discourse, the strongest link is between “Space race & rivalry” and “Spaceflight & missions,” where competitive framing frequently co-occurs with references to operational capability and concrete achievements.

Figure 3. Aggregate theme distribution and co-occurrence view in the New York Times corpus

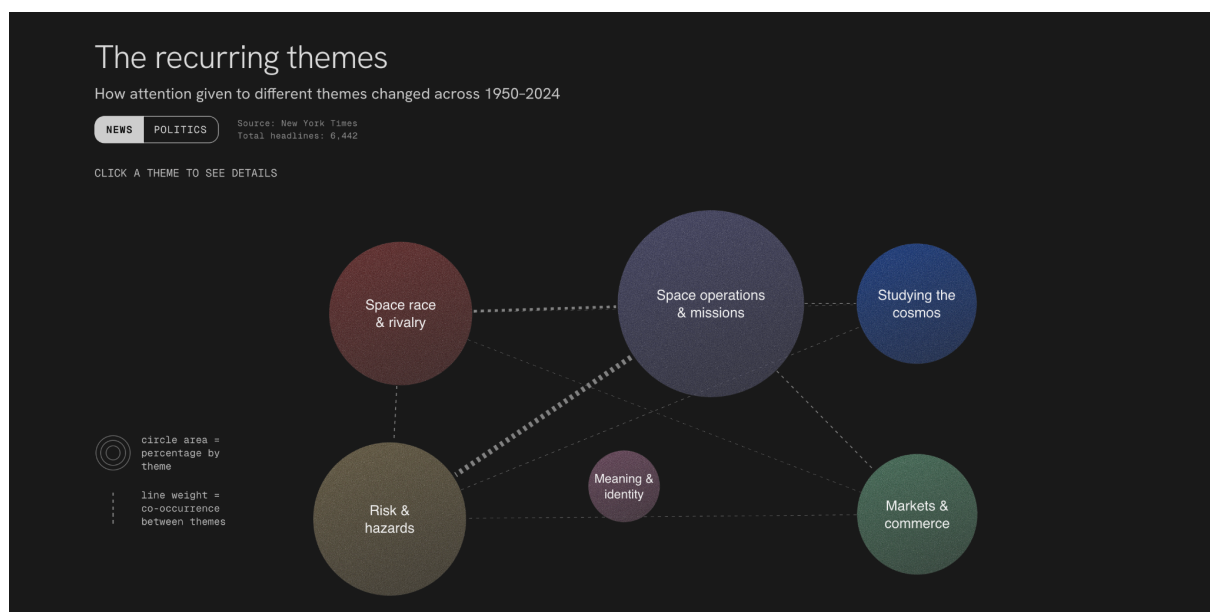
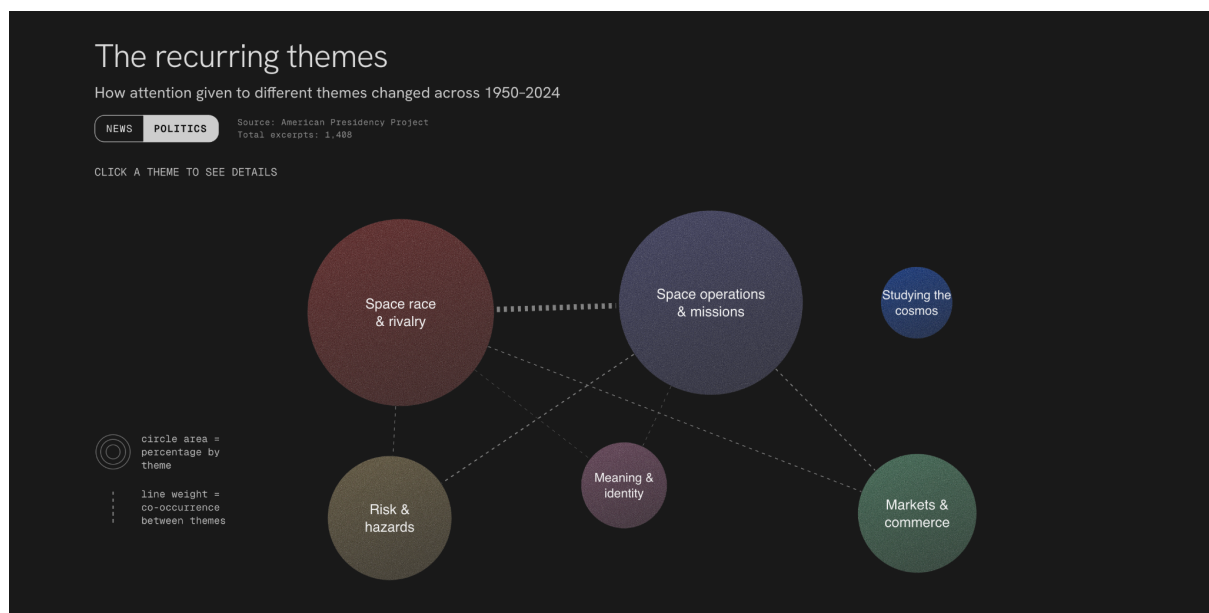


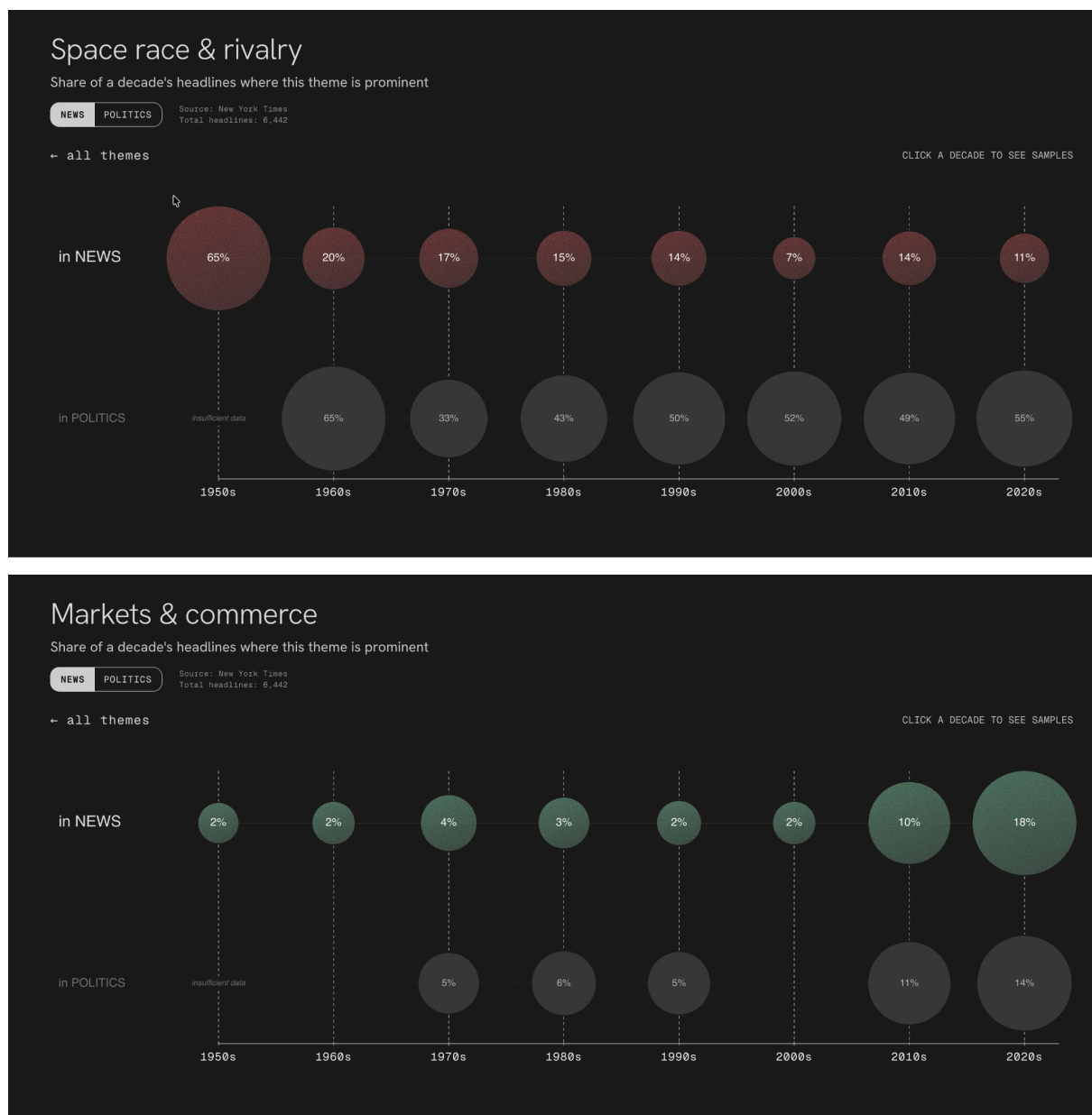
Figure 4. Aggregate theme distribution and co-occurrence view in the presidential corpus



Selecting a theme transitions to a decade-level view that tracks its prominence over time. In this representation, each document has a clearly defined top theme. A theme is considered prominent if its softmax weight falls within 15% of that top-scoring theme. This ensures that the dominant theme is always included, while also capturing cases where one or more additional themes are nearly as strong. Because documents may be prominent in multiple themes, the proportions across themes within a decade can exceed 100%, reflecting overlapping narratives.

This view reveals distinct temporal patterns. “Space race & rivalry” declines sharply in news after the Cold War but remains prominent in political discourse. “Markets & commerce” shows a rapid increase in both corpora in the last decades, reflecting the emergence of commercial space actors. “Meaning & identity” appears only intermittently, primarily in presidential speech during the 1960s and 1970s. “Spaceflight & missions” remains consistently dominant across corpora, reflecting the persistent public focus on the physical execution of space activity.

Figure 4. Decade-level theme prominence view



2.4.2. Talking To Space

The *Talking to Space* section presents a complementary dataset: a catalogue of cultural artifacts and messages sent beyond Earth. The visualization is structured as a radial timeline, with each category represented as a concentric arc and each object plotted according to its year.

The radial format emphasizes changes in density over time. A secondary, linear timeline is available for decade-level exploration, allowing users to examine dense periods in more detail. Together, these views balance overview and granularity.

The catalogue reveals a transformation in how humanity addresses space. This shift is not only quantitative but structural: the nature of the objects changes from mainly institutionally curated messages to a diverse set of contributions from private actors and individuals. This transition parallels the shift toward commercialization observed in the discourse analysis, suggesting that the same underlying forces shaping how space is discussed are also shaping how it is engaged with materially.

Figure 5. Radial timeline of cultural artifacts and messages sent to space

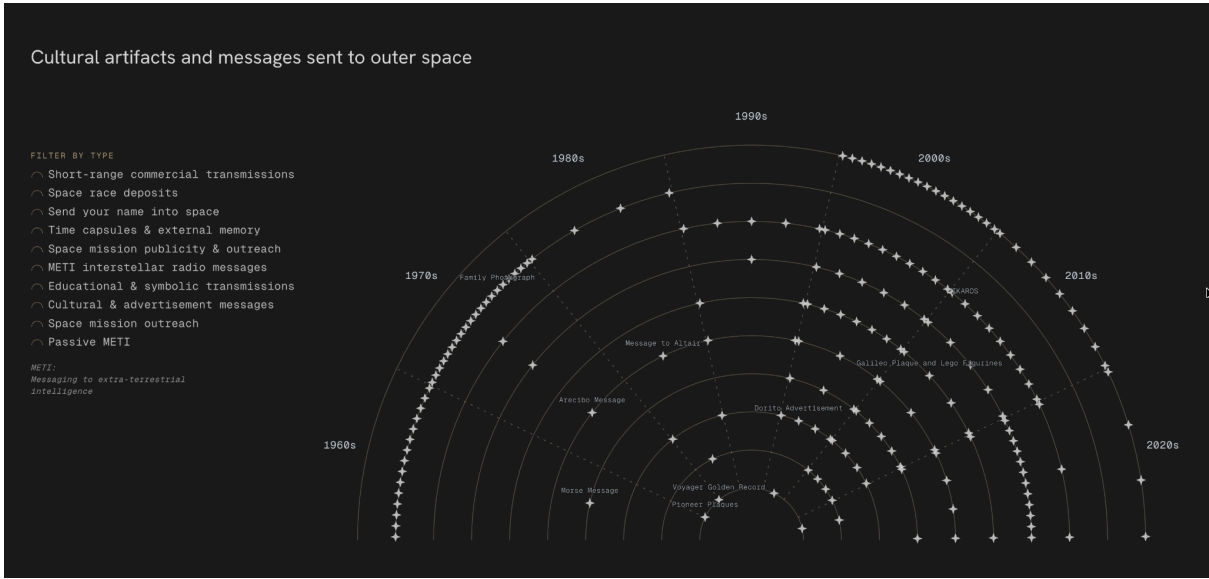
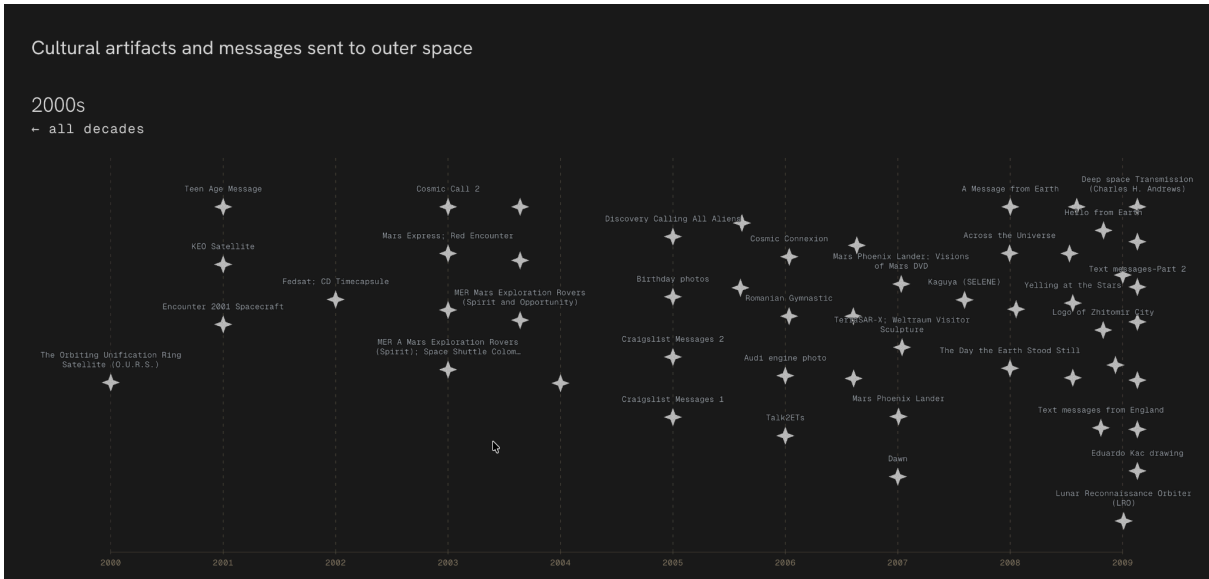


Figure 6. Decade-level timeline of cultural artifacts and messages sent to space



2.5. Limitations

Several limitations shape the interpretation of this project. First, the corpora are not exhaustive records of space discourse. The New York Times is used as a proxy for elite mainstream media, while the American Presidency Project captures presidential communication rather than political rhetoric as a whole.

Second, the keyword-based retrieval strategy prioritizes precision over recall: it ensures that analyzed documents are explicitly space-related, but it also means that the search vocabulary partly shapes the thematic structure of the corpus.

Third, the embedding-based classification should be interpreted as a system of relative thematic and actor prominence, not as calibrated probability or ground-truth labeling.

Fourth, the Talking to Space catalogue is treated as a curated qualitative dataset rather than as an exhaustive or representative population of all symbolic objects and messages sent beyond Earth.

These limitations do not undermine the comparative patterns surfaced by the project, but they define the scale at which the findings should be read: as visual evidence of broad narrative structures and shifts, rather than as exhaustive measurement of all public discourse about space.

3. Conclusion

This project set out to examine how outer space has been publicly constructed in the United States across seven decades, not as a physical domain alone, but as a field of meaning shaped through language. By combining large-scale text analysis with interactive visualization, it traces how space has been described, who has been placed at its center, and how those narratives have shifted over time.

Several patterns emerge. First, space discourse is not uniform across institutional contexts. Elite mainstream news coverage and presidential rhetoric organize the same domain differently: they emphasize different actors, privilege different themes, and encode distinct narrative logics.

Second, certain narrative structures persist even as the underlying context changes. For instance, the rivalry frame does not disappear from politics with the end of the Cold War but is reconfigured around new actors and technologies, shifting from the Soviet Union to contemporary rival countries and from symbolic milestones to operational capabilities such as satellites and anti-satellite systems.

Third, the evolution of actors reveals a broader transformation in how space activity is socially understood. The central figure of the astronaut during the Apollo era—representing collective, state-mediated achievement—gives way to institutional actors and, more recently, to private-sector figures. This transition parallels the rise of “Markets & commerce” as a dominant theme.

The *Talking to Space* catalogue provides a complementary perspective on these changes. The shift from carefully curated, institutionally produced messages—such as the Pioneer plaques and the Voyager Golden Record—to a more fragmented and commercially driven set of contributions reflects the same underlying transformation.

Beyond these findings, the project also advances a methodological approach. It translates a qualitative question—how has space been framed?—into a comparative visual analysis of themes, actors, and symbolic projections. By combining corpus construction, embedding-based classification, and interactive visualization, it offers a framework for large-scale discourse analysis that is both scalable and interpretable. This allows the analysis to remain transparent and understandable, rather than a black-box prediction. The value of this approach lies in making analytical decisions explicit and reproducible, allowing patterns that would be difficult to observe qualitatively to be examined at scale.

Ultimately, the project shows that outer space is not only explored but also continuously constructed through narration. Across decades, the story of space remains a contested and evolving domain, shaped by the institutions that describe it and the actors that engage with it.

Understanding space, in this sense, requires not only tracking rockets and satellites, but also tracing the language through which space is imagined, framed, and made meaningful.

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ANNEX A

Search terms used to query the NYT Article Search API (headline filter) and the American Presidency Project (full-text search)

space race, space age, space program, space exploration, space policy, space treaty, space station, space shuttle, astronauts, NASA, Apollo 11, moon landing, Sputnik, SpaceX, Blue Origin, space tourism, Starlink, Starship, dark matter, alien life, extraterrestrial, life on mars, weapons in space, space weapons, anti-satellite

Note: The keyword “Sputnik” generated a small number of false positives referencing the Sputnik V COVID-19 vaccine. These were identified and removed manually after collection.

ANNEX B

Theme Anchor Queries

Space race & rivalry

geopolitical competition: Soviet space program threat to United States, USSR space achievement beats America, Cold War space competition between superpowers, Soviet satellite ahead of American, Soviet rocket surpasses American capability, catching up to Soviet Union in space, American response to Soviet space milestone, US falling behind in space race, Soviet cosmonaut ahead of American astronaut, race between US and USSR in space, Russian military satellite threatens United States, Russian anti-satellite weapon test, China space program challenges American dominance, Chinese military satellite capability, Chinese anti-satellite test threatens US, rival nation space weapon program, space arms race between nations, military competition in space, US military space supremacy challenged by foreign power, space race between rival nations

Spaceflight & missions

physical space operation: spacecraft in orbit, crewed mission to orbit, astronaut spacewalk, rocket in flight, satellite deployed, space station crew aboard, probe reaches planet, rover lands on Mars, lander touches down, orbital insertion burn completed, rendezvous and docking in orbit, telemetry signal received, crew splashdown, capsule returns to Earth, Earth observation satellite operational, remote sensing mission active, physical spacecraft activity in orbit, NASA budget allocation for space missions

Studying the cosmos

scientific measurement: telescope detects, astronomers observe, spectroscopic analysis reveals, orbital period measured, cosmic radiation detected, galaxy formation observed, black hole imaged, gravitational wave detected, dark matter evidence, cosmic microwave background, stellar evolution observed, telescope observation data, spectroscopic measurement of atmosphere, orbital data analysis, cosmic signal detection, astronomical observation results

Risk & hazards

accident event: rocket explosion, launch failure, spacecraft crash, catastrophic malfunction, crew killed in space, astronauts dead, disaster investigation findings, Columbia disaster, Challenger explosion, Apollo fire tragedy, space debris collision risk, Kessler syndrome, orbital collision danger, radiation

exposure hazard in space, solar storm threatens satellites, life support failure, oxygen depletion emergency, crew emergency in space, stranded astronauts, rescue mission, communication blackout emergency, shuttle disaster cause, mission failure investigation, explosion event, crash incident, fatal accident

Markets & commerce

private sector activity: SpaceX, Blue Origin, Virgin Galactic, Rocket Lab, space entrepreneur, private space company, commercial spaceflight, space tourism, private astronaut mission, commercial satellite operator, launch market competition, reusable rocket business, billionaire space race, private sector leading space, commercialization of space, private investment in space, space industry led by companies, commercial space industry, private launch provider, satellite internet service, public private partnership in space, commercial launch activity, Elon Musk space venture, Jeff Bezos rocket company

Meaning & identity

philosophical meaning: why we go to space, purpose of space exploration, meaning of moon landing, space as human destiny, overview effect, pale blue dot, cosmic perspective, what space means for humanity, existential significance of spaceflight, legacy of Apollo for civilization, space age as cultural turning point, pioneering spirit of space exploration, space as next frontier for humanity, are we alone in universe, Fermi paradox, cosmic loneliness, space colonialism debate, who should own space, humanity's place in cosmos, philosophical meaning of reaching the moon, space as mirror for society, what the moon landing meant for the world, space age doubt, rethinking the space program's purpose

ANNEX C

Actor Anchor Queries

Astronauts & cosmonauts

Regex-controlled: \b(astronauts?|cosmonauts?)\b

National state

NASA budget appropriation Congress, federal space funding vote, NASA administrator appointed fired, White House space policy directive, NASA program canceled delayed approved, Congressional space hearing, Space Force Pentagon military space, Department of Defense satellite, NASA contract awarded federal agency, federal space agency management, NASA workforce layoffs hiring freeze, NASA headquarters decision, presidential executive order space, NASA inspector general audit, NASA safety review board investigation, Senate House NASA authorization bill, NASA administrator testimony Congress oversight, NASA restructuring reform reorganization, government space agency leadership, NASA civil service workforce discrimination, federal space budget cut increase, NASA scientists discover find publish, NASA research mission findings, U.S. space agency announces result, NASA mission reveals data

Geopolitical rivals

Soviet Gagarin Tereshkova Leonov orbits Earth, USSR Sputnik satellite launched orbit Moscow announces, Vostok Voskhod Soviet spacecraft launches, Soviet space station Salyut docks mission, Soyuz spacecraft Soviet orbit, Soviet space disaster accident, Roscosmos Russia launches rocket satellite payload, Russian mission docks station orbit, Russian anti-satellite weapon ASAT test destroys satellite, Russian nuclear space weapon deployment threat, Russia military satellite reconnaissance program, China CNSA Shenzhou launches orbit, Tiangong Chinese space station docks, Chang'e lunar lander China Moon surface lands, Tianwen Mars probe Beijing launches, China destroys satellite debris ASAT test, Chinese military space PLA capability announced, Beijing space mission achievement announces, Russia China space weapons adversary threat capability, adversary anti-satellite weapon threatens US space, Moscow Kremlin space program strategic challenge

International partnerships

European Space Agency ESA Ariane mission program launches, JAXA Japan Aerospace Exploration Agency spacecraft rocket, ISRO Indian

Space Research Organisation lunar Chandrayaan mission, Canadian Space Agency CSA Canadarm contribution, Artemis Accords nation signs commits framework, Outer Space Treaty multilateral obligation ratified, ESA JAXA ISRO CSA named partner agency agreement, foreign space agency bilateral framework signed, United Nations outer space committee COPUOS resolution, Arianespace European commercial launch provider

Private sector

SpaceX valuation funding billion investors, Blue Origin investment capital raises, Virgin Galactic stock IPO shares trading, Rocket Lab revenue earnings market, private space company layoffs workforce cuts, space startup venture capital funding, commercial launch market competition provider, Starlink broadband service subscribers pricing revenue, space tourism ticket price passenger paying, private rocket company profit loss quarterly, space company merger acquisition deal, NewSpace industry private capital investment, space economy market growth private sector, SpaceX valuation privately held company worth, Elon Musk SpaceX company builds tests launches, Jeff Bezos Blue Origin rocket company, Richard Branson Virgin Galactic flight, SpaceX Starship rocket test flight explosion, Bezos Musk billionaire rocket company competition, private space company achieves milestone first, SpaceX reusable rocket lands recovery, commercial crew capsule private company, NewSpace company founder vision private, private rocket firm independent of government, commercial space station private company builds, space tourism private passenger flight civilian

Scientific community

astronomers discover detect announce findings published, astrophysicists measure evidence data analysis journal, researchers publish peer reviewed study results, university observatory telescope survey detects, science team outside government reports evidence, research institute announces discovery space phenomenon, physicists cosmologists detect dark matter evidence, exoplanet discovered astronomers telescope survey, black hole imaged observed event horizon telescope, independent scientists find evidence conclude, academic study astrophysics journal paper publishes, telescope array detects signal researchers announce, scientific journal Nature Science publishes space finding, named scientist professor doctor research group discovers, science community debates challenges replication findings

ANNEX D

Pairwise cosine similarity between theme anchor embeddings

	Studying the cosmos	Spaceflight & missions	Space race & rivalry	Risk & hazards	Markets & commerce	Meaning & identity
Studying the cosmos	1.0000	0.7034	0.6312	0.6738	0.6003	0.6468
Spaceflight & missions	0.7034	1.0000	0.7239	0.7689	0.7298	0.7065
Space race & rivalry	0.6312	0.7239	1.0000	0.6755	0.6708	0.6261
Risk & hazards	0.6738	0.7689	0.6755	1.0000	0.6749	0.6546
Markets & commerce	0.6003	0.7298	0.6708	0.6749	1.0000	0.6168
Meaning & identity	0.6468	0.7065	0.6261	0.6546	0.6168	1.0000

Pairwise cosine similarity between actor anchor embeddings

	Astronauts	National state	Geopolitical rivals	Int. partners	Private sector	Scientific community
Astronauts	1.0000	0.5676	0.6096	0.6179	0.5909	0.5638
National state	0.5676	1.0000	0.6567	0.6883	0.6720	0.6755
Geopolitical rivals in space	0.6096	0.6567	1.0000	0.7005	0.6349	0.6116
Int. partners	0.6179	0.6883	0.7005	1.0000	0.6771	0.6208
Private sector	0.5909	0.6720	0.6349	0.6771	1.0000	0.5912
Scientific community	0.5638	0.6755	0.6116	0.6208	0.5912	1.0000